

A Visual Guide to Reproduction in Flowering Plants

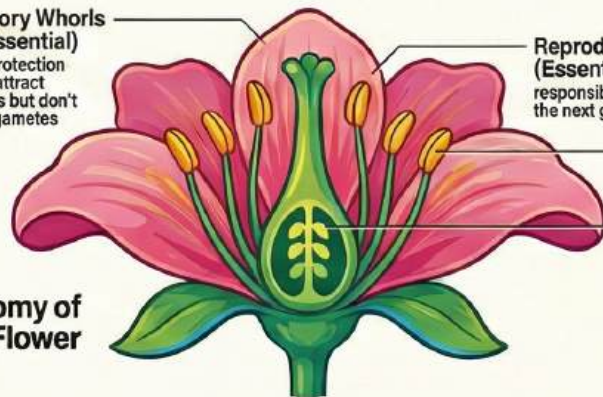
Accessory Whorls (Non-Essential)
provide protection and help attract pollinators but don't produce gametes

Reproductive Whorls (Essential)
responsible for creating the next generation

Androecium (Male Part)
Stamens

Gynoecium (Female Part)
Pistile

Anatomy of a Flower



Anther

Pre-Fertilization: Creating the Gametes

Male Gamete Formation (in the Stamen)

Pollen Mother Cell (2n)

Meiosis

Haploid (n) Microspores

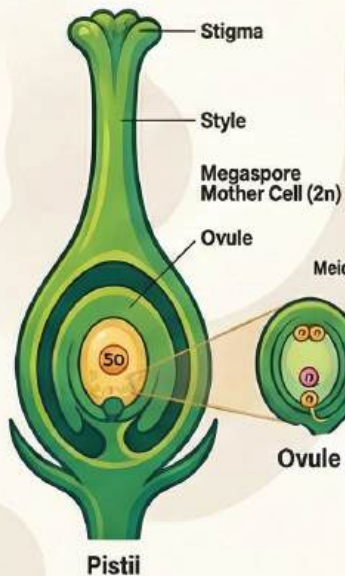
Stamen

The Pollen Grain is the Male Gametophyte. This structure is responsible for producing the male gametes

Special Reproductive Modes

Apomixis:
female without fertilization (asexual reproduction, no gamete fusion)

Polyembryony:
The presence of more than one embryo in a single seed.



Pistil

Two Main Types: Self-Pollination and Cross-Pollination

Agents of pollination can be abiotic (wind, water) or biotic (insects, birds, bats)

Pollination: Bridging the Gap

Pollination is the transfer of pollen from anther to stigma. This process is essential for bringing the male and female gametes together for fertilization

Double Fertilization: The Core Event

Two separate fusion events occur simultaneously. This unique process ensures the formation of both the embryo and its food source

Fusion 1: Syngamy

One Male Gamete (n)

Female Gamete (n)

Fusion 2: Triple Fusion

One Male Gamete (n)

Zygote (2n)

Polar Nuclei (nn)

Primary Endosperm Nucleus (3n)

Post-Fertilization: Development of Seed and Fruit

Embryo
(future plant through embryogeny)

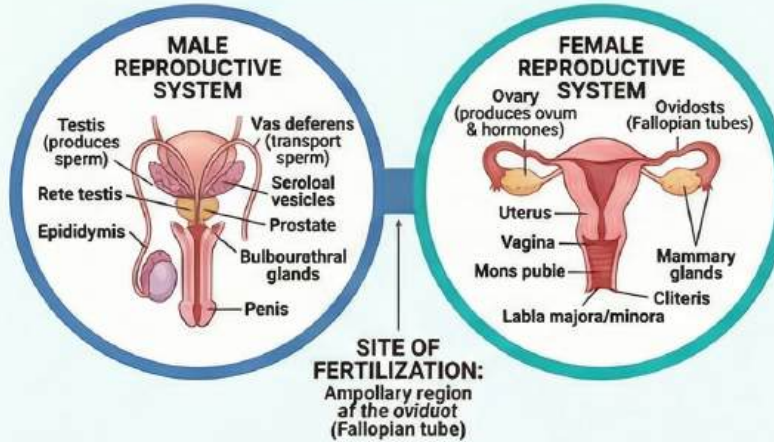
Endosperm
(triploid nutritive tissue)

Seed
(Contains embryo and endosperm)

Fruit
(protecting the seed)
@Thunderstudy

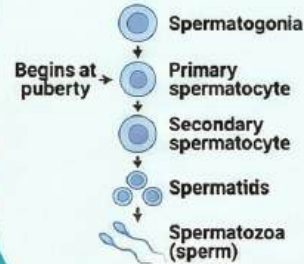
The Journey of Human Reproduction: A Quick Revision Guide

THE REPRODUCTIVE SYSTEMS

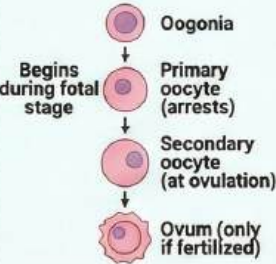


GAMETOGENESIS: Creating Sperm & Egg

SPERMATOGENESIS (in Males)



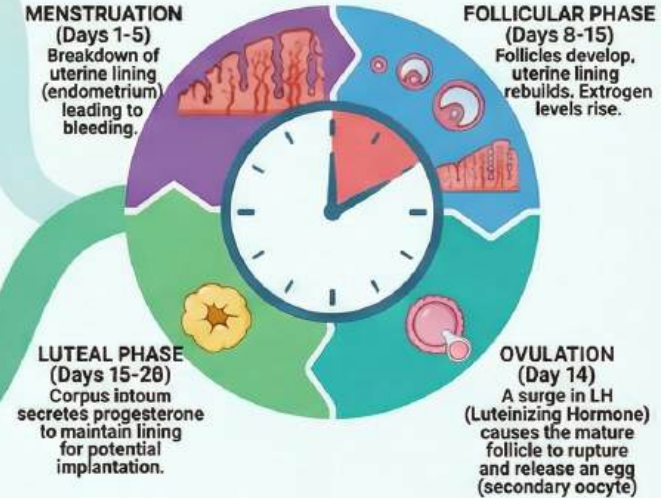
OOGENESIS (in Females)



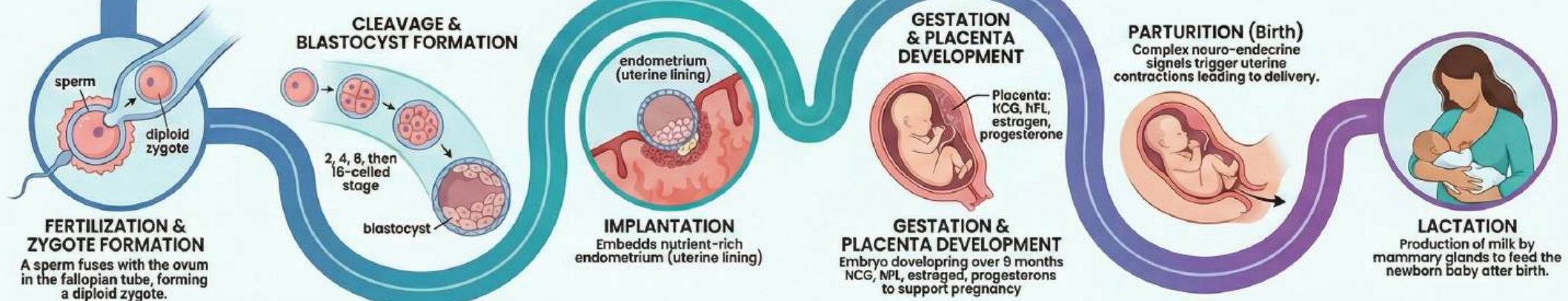
CRITICAL TIMING DIFFERENCE:

Spermatogenesis is a continuous process starting from puberty, while Oogenesis starts before birth, pauses, and completes a cycle monthly after puberty.

THE MENSTRUAL CYCLE (28/29-Day Cycle)



FROM FERTILIZATION TO BIRTH



Reproductive Health: A Quick Revision Guide

Birth Control Methods (Contraception)



Natural Methods

Avoid meeting of sperm and ovum without devices/pills. No side effects, high failure rate.

Coitus Interruptus (Withdrawal): The withdrawal of the penis from the vagina just before ejaculation.

Lactational Amenorrhea: The temporary absence of menstruation for up to 6 months after childbirth due to intense breastfeeding, which prevents ovulation.



Barrier Methods

Physically block sperm from reaching ovum. Can use with spermicides.

Examples:

Condoms (Male & Female) - "Nirodh" is a popular indian brand; **Diaphragms, Cervical Caps.**



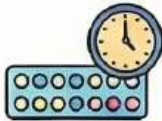
Intrauterine Devices (IUDs)

Inserted by medical experts into the uterus to prevent pregnancy.

Non-Medicated (Lippes Loop),

Copper Releasing (Copper T, Copper 7, Multiload 375).

Hormone Releasing (Progestosert, LNG-20).



Oral Contraceptives (Pills)

Daily/weekly hormonal pills inhibit ovulation and implantation.

Two Main Types: Combination Progestogen-Estrogen or Progestogen-only.

Example: 'Saheli' (Non-steroidal, once-a-week, developed in India).



Surgical Methods (Sterilization)

Permanent and irreversible.

Vasectomy (in Males): Vas deferens cut/tied.

Tubectomy (in Females): Fallopian tubes cut/tied.



A State of Total Well-being

According to the WHO, reproductive health means complete physical, physical, mental, and social well-being in all aspects of reproduction.

Key Medical Procedures & Issues



Amniocentesis

Prenatal test analyzing amniotic fluid for genetic disorders.

Misuse leads to Female Feticide: Test illegally used to abort female fetuses.



Medical Termination of Pregnancy (MTP)

Intentional/voluntary termination (induced abortion).

Legally Permitted Cases: Unwanted pregnancies (rape, contraceptive failure), or harmful to mother/letus.

Safety Window: Relatively safe during first trimester (up to 12 weeks), riskier thereafter.



Assisted Reproductive Technologies (ART)

Helping Infertile Couples Conceive: Medical procedures to treat infertility.

Examples:

ZIFT (Zygote intrafallopian Transfer): Zygote from lab transferred to fallopian tube.

GIFT (Gamete intrafallopian Transfer): Donor ovum transferred to fallopian tube.

ICSI (Intracytoplasmic Sperm Injection): Single sperm directly injected into ovum in lab.

AI (Artificial Insemination): Semen from male partner/donor introduced into female reproductive tract.



Sexually Transmitted Diseases (STDs)

What are STDs?

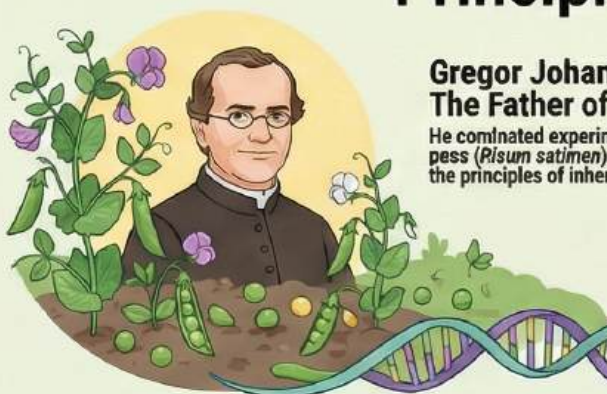
Infections/diseases transmitted through sexual intercourse.

Also known as STIs or VD.

Common STDs:

Gonorrhoea, Syphilis, Genital Herpes, Hepatitis-B, HIV/AIDS.

Principles of Inheritance & Variation: A Visual Guide



Gregor Johann Mendel: The Father of Genetics

He conducted experiments on garden peas (*Pisum sativum*) to understand the principles of inheritance.

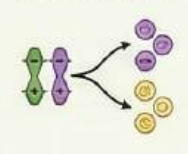
Mendel's Three Laws of Inheritance

Law of Dominance



In a hybrid, only the dominant allele expresses itself in the F1 generation.

Law of Segregation



During gamete formation, the two alleles for a trait separate (segregate) from each other.

Law of Independent Assortment



Alleles for different traits assort independently of one another during gamete formation.

Proving the Laws: The first two laws were demonstrated using a monohybrid cross, while the third law was proven with a dihybrid cross.

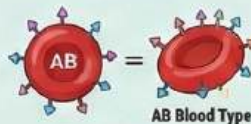
Mendel's Seven Characters

Stem Height	Yell (Dominant)	Purpl (Recessive)	Pod Shape	Inflated (Dominant)	Constricted (Recessive)
Flower Color	Violet (Dominant)	White (Recessive)	Pod Color	Green (Dominant)	Yellow (Recessive)
Seed Shape	Axial (Dominant)	Terminal (Recessive)	Flower Position	Axial (Dominant)	Terminal (Recessive)

Beyond Mendel: Non-Mendelian Inheritance



Codominance

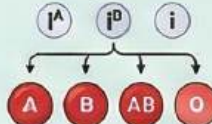


Both dominant and recessive alleles express themselves fully and equally in the F1 generation.

Incomplete Dominance

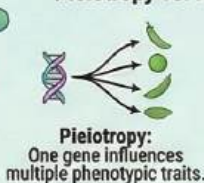
The dominant allele does not completely mask the recessive allele, resulting in a blended, intermediate phenotype.

Multiple Alleles

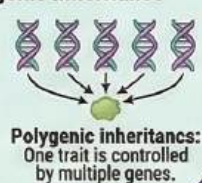


A gene that exists in more than two allele forms within a population.

Pleiotropy vs. Polygenic Inheritance

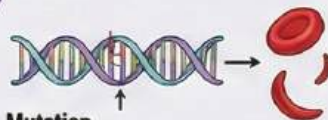


Pleiotropy: One gene influences multiple phenotypic traits.



Polygenic inheritance: One trait is controlled by multiple genes.

Core Concepts in Genetics

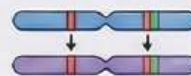


Mutation

A sudden, heritable change in the DNA sequence which can alter the genotype and phenotype of an organism. Point mutation is a key type, exemplified by sickle-cell anemia.

Chromosomal Theory of Inheritance

Proposed by Sutton and Baveri in 1902, it states that genes are located on chromosomes. It was experimentally verified by Thomas Hunt Morgan using fruit flies (*Drosophila melanogaster*).



Linkage

The term, coined by Morgan, describes the tendency of genes located close together on the same chromosome to be inherited together.

Genetic Disorders & Analysis

Two Main Types of Genetic Disorders

Mendelian Disorders

Caused by alteration in a single gene



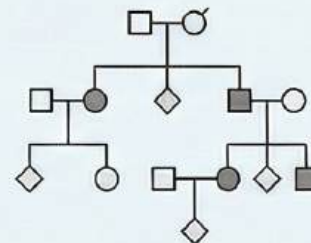
Chromosomal Disorders

Caused by absence, excess, or abnormal arrangement of chromosomes



Pedigree Analysis

The study of a particular trait's inheritance through a family's generations to determine how it is passed on.



Pedigree Symbols

- Male
- Female
- ◇ Sex Unspecified
- Affected Individual
- Mating
- == Consanguineous mating (between relatives)

Chromosomal Sex Determination Mechanisms

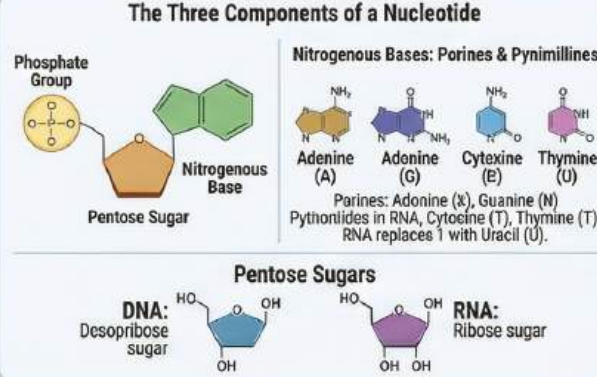
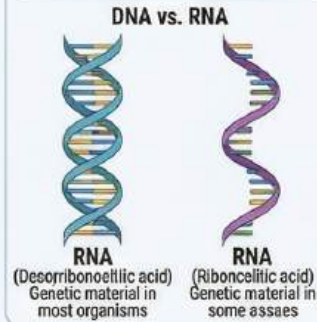
Humans XX-XY Type	Birds ZZ-ZW Type	Grasshopper XX-XO Type
Female: XX	XX ZW	XX
Male: XY	ZW ZZ	XX XO
Female: XX	ZW ZW	XX
Male: XY	ZW ZZ	XO

Decoding the Blueprint of Life: A Quick Guide to the Molecular Basis of Inheritance

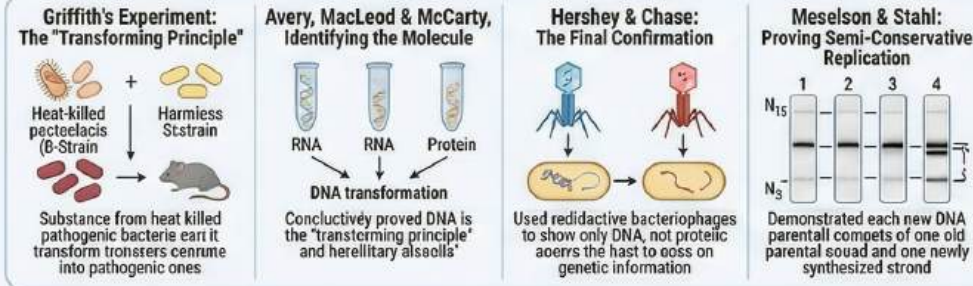
THE BUILDING BLOCKS OF GENETIC MATERIAL

Nucleic Acid: The Genetic Material

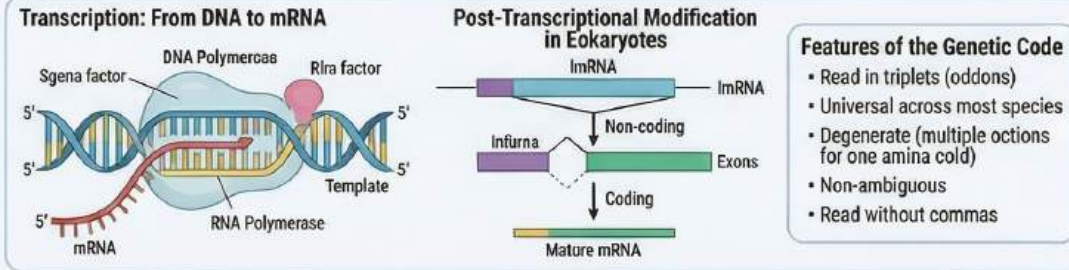
Primary carriers of genetic information



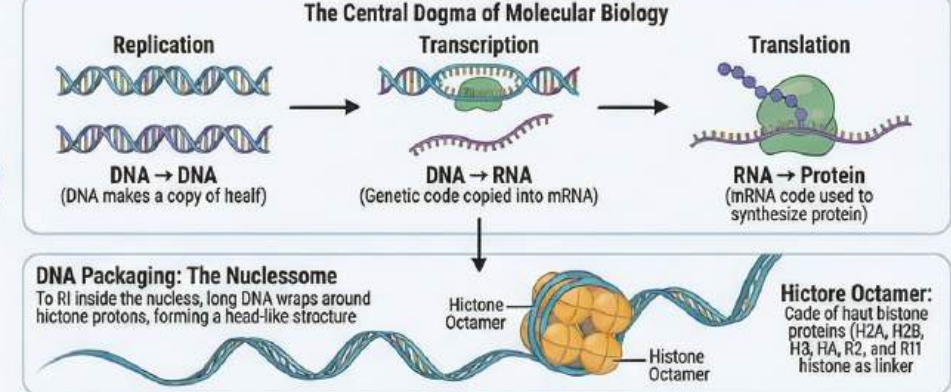
LANDMARK EXPERIMENTS IN GENETICS



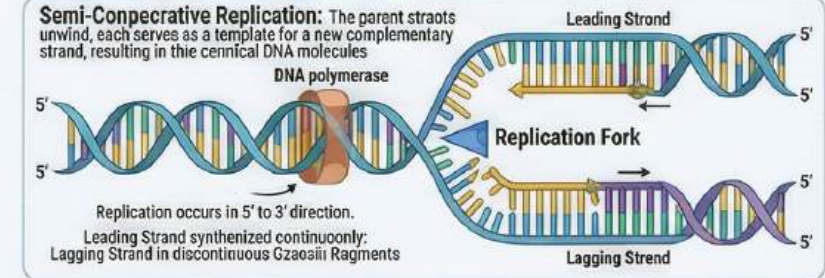
FROM GENE TO PROTEIN: EXPRESSION



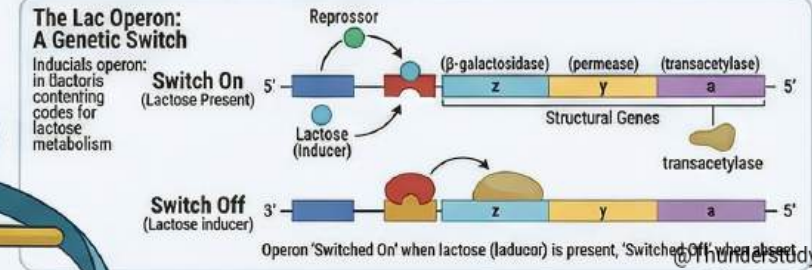
THE CENTRAL DOGMA: INFORMATION FLOW IN THE CELL



DNA REPLICATION: COPYING THE CODE



GENE REGULATION: THE LAC OPERON



Evolution: A Quick Revision Guide

EVIDENCE FOR EVOLUTION

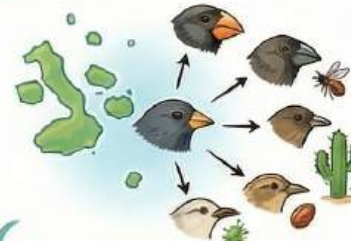
PALEONTOLOGICAL (FOSSIL EVIDENCE)

Fossils reveal extinct animals, connecting links, and evolutionary history (phylogeny).

HOMOLOGOUS VS. ANALOGOUS ORGANS

HOMOLOGOUS ORGANS	ANALOGOUS ORGANS
Some Internal Structure Different Functions Divergent Evolution	Different Internal Structure, Same Function Convergent Evolution
	
Examples: Perilimbs of Human, Whale, Bat, Chostair.	Examples: Wings of Birds & Insects; Eyes of Odogus & Mammals.
Plant Examples: Thorns of Bougalsibes & Tamills of Cacorchila.	Plant Examples: Sweet Potato (Root) & Potato (Stum)

BIOGEOGRAPHICAL (ADAPTIVE RADIATION)



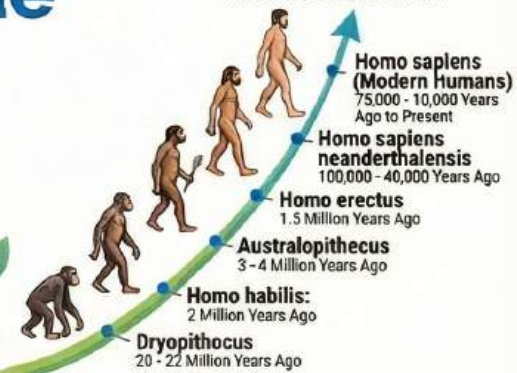
Darwin's Finches: On the Galapagos Islands, an original seed-eating finch species evolved into many different varieties with unique beak shapes adapted for eating insects, fruits, or cactus.

CONVERGENT EVOLUTION



Placental lead to similar forms (e.g., Placents & Australian Marsupials).

A TIMELINE OF HUMAN EVOLUTION



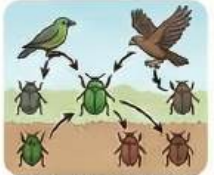
MECHANISMS & THEORIES OF EVOLUTION

LAMARCKISM: USE AND DISUSE



Organisms acquire traits through use/disuse, passed to offspring.

DARWINISM: NATURAL SELECTION



Nature 'selects' individuals with traits suited for survival ('survival of the fittest').

HARDY-WEINBERG PRINCIPLE

$$p^2 + 2pq + q^2 = 1$$

Describes genetic equilibrium in a population (alleles frequencies constant).

5 FACTORS THAT DISRUPT EQUILIBRIUM (CAUSE EVOLUTION)

1. **Gene flow** between populations
Arrows between populations
2. **Genetic Drift**
random population change
3. **Mutation**
DNA change
4. **Genetic Recombination**
Shuffling genes
5. **Natural Selection**
selective pressure

THEORIES ON THE ORIGIN OF LIFE



Theory of Panspermia
Life came to Earth from outer space in the form of spores.



Theory of Spontaneous Generation
Life arose from non-living matter spontaneously.

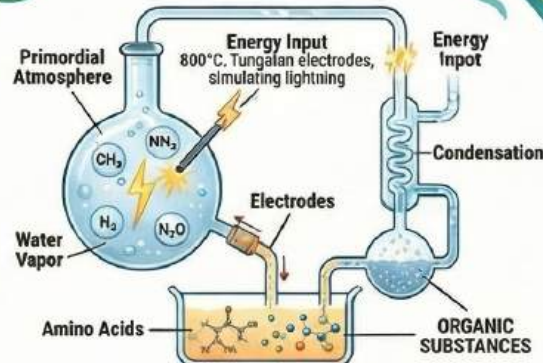


Theory of Biogenesis
Life originates only from pre-existing life.



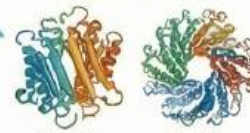
Theory of Chemical Evolution
Life originated from inorganic substances through chemical reactions on early Earth (Oparin & Haldans).

THE MILLER-UREY EXPERIMENT: PROOF OF CHEMICAL EVOLUTION



An Experiment to Simulate Early Earth (1953).
They created a closed system with reducing atmosphere to mimic lightning.
Result. After a week, amino acids and organic substances formed.

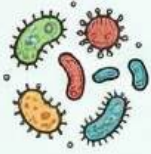
BIOCHEMICAL & EMBRYOLOGICAL EVIDENCE



Proteins
Similarities in proteins, genes, and early embryonic stages point to shared ancestry.



Human Health & Disease: A Quick Revision Guide



What Causes Disease?

Pathogens are disease-causing organisms. These include bacteria, viruses, fungi, protozoa, and helminths (worms).

Common Diseases & Their Pathogens

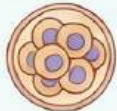
Disease	Pathogen Type	Causal Organism
Typhoid	Bacterial	<i>Salmonella typhi</i>
Pneumonia	Bacterial	<i>Streptococcus pneumoniae</i>
Common Cold	Viral	Rhinovirus
Malaria	Protozoan	<i>Plasmodium</i> (e.g., <i>P. falciparum</i> , <i>P. vivax</i>)
Amoebiasis	Protozoan	<i>Entamoeba histolytica</i>
Ascariasis	Helminth (Roundworm)	<i>Ascaris</i>
Filariasis (Elephantiasis)	Helminth (Filarial worm)	<i>Wuchereria bancrofti</i>



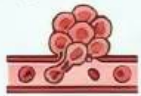
Key Finding: Typhoid is diagnosed using the Widal Test. This bacterial disease is caused by *Salmonella typhi*.

Understanding Cancer

Benign vs. Malignant Tumors



Benign:
Contained, no spread



Malignant:
Cancerous, spread to other body parts (metastasis)

Detection Methods



Biopsy, Radiography (X-ray), CT scans, MRI

Treatment Options



Surgery, Radiation Therapy, Chemotherapy, immunotherapy (often combined)

What is True Health?



Health is a state of complete physical, mental, and social well-being, not merely the absence of disease.

Our Body's Defense: The Immune System

Innate Immunity (By Birth & Non-Specific)



Physical:
Skin, mucus membranes



Physiological:
Stomach acid (HCl), saliva, tears



Cellular:
White Blood Cells (e.g., Neutrophils)



Cytokine:
Interferons that protect non-infected cells from viruses.

Acquired Immunity (Learned & Specific)



Active Immunity:
Body produces own antibodies after infection/vaccination



Passive Immunity:
Ready-made antibodies received (e.g., Colostrum, anti-venom)



Antibody Structure
H2L2: Two heavy chains and two light chains held by disulfide bonds

Key Players: B and T Lymphocytes.

T cells help identify pathogen; B cells produce an army of specific antibodies.

When Immunity Goes Wrong



Allergy

An exaggerated immune response to certain substances called allergens. The antibody involved with anti-histamines.



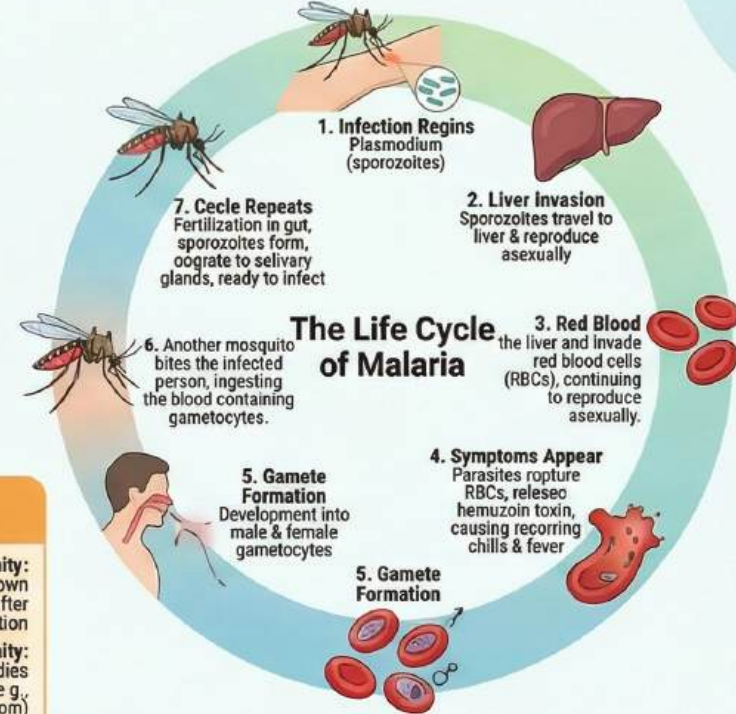
Autoimmunity

Immune system loses ability to differentiate foreign cells from 'self', attacking own tissues (e.g., Rheumatoid Arthritis).

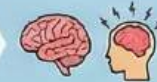


AIDS (Acquired Immunodeficiency Syndrome)

Caused by HIV retrovirus, destroys helper T-cells, severely weakening the immune system, making vulnerable to infections.



Drug & Alcohol Abuse



Common Causes

Curiosity, peer pressure, stress, academic pressure, escape from problems.



Warning Signs

Sudden drop in academic performance, isolation, depression, loss of interest, lack of hygiene.



Prevention & Control

Education and counseling, seeking help from parents/peers, avoiding undue peer pressure.

@Hunderstudy

The Mighty Microbes: Our Tiny Allies in Daily Life

Microbes in the Home



Curd (Yogurt) Production

Lactobacillus bacteria partially digest and coagulate milk proteins to turn milk into curd. The starter culture is called an inoculum.



Fermented Dough

Yeast (*Saccharomyces cerevisiae*) ferments sugars in dough, producing Carbon Dioxide and become spongy for bread, idli, and dosa.



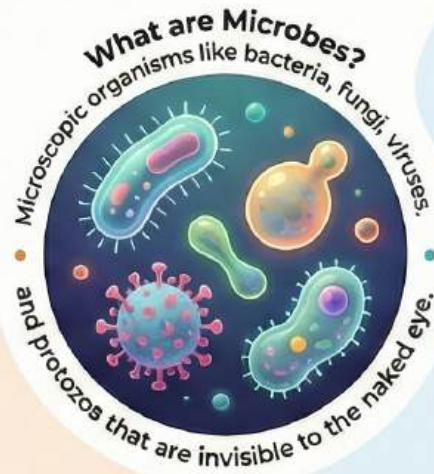
Swiss Cheese

The bacterium *Propionibacterium shermanii* produces large amounts of CO_2 during fermentation, creating characteristic large holes in Swiss cheese.



Toddy

A traditional South Indian drink made by fermenting the sap from palm trees.



Microbes in Industry



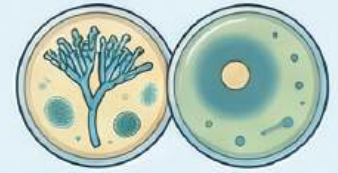
Fermented Beverages

Yeast (*Saccharomyces cerevisiae*, Brewer's Yeast) ferments malted cereals and fruit juices to produce alcoholic beverages like ethanol.



Production of Organic Acids

Citric Acid	<i>Aspergillus niger</i> (Fungus)
Acetic Acid	<i>Acetobacter aceti</i> (Bacterium)
Butyric Acid	<i>Clostridium butylicum</i> (Bacterium)
Lactic Acid	<i>Lactobacillus</i> (Bacterium)



The First Antibiotic: Penicillin

Discovered by chance, Penicillin is produced by the fungus *Penicillium notatum*.



Enzyme Production

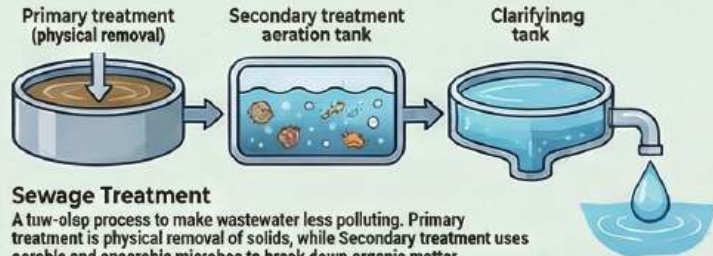
Lipases
Pectinases & Proteases
Streptokinase (Clot Buster)
<i>Streptococcus</i> (Bacterium)



Bioactive Molecules

Statins (Cholesterol-lowering)
<i>Monascus purpureus</i> (Fungus)
Cyclosporin A (Immunosuppressant)
<i>Trichoderma polysporum</i> (Fungus)

Microbes in Environmental Management



Biogas (Gobar Gas) Production

Anaerobic bacteria called Methanogens (*Methanobacterium*) digest cattle dung and other organic wastes to produce a mixture of gases, primarily methane (CH_4), CO_2 , and H_2S .



Biocontrol Agents: Nature's Pesticides

Using natural predators of pests to control them, avoiding harmful chemical pesticides. Examples include *Bacillus thuringiensis* for butterfly caterpillars and *Trichoderma* fungus.



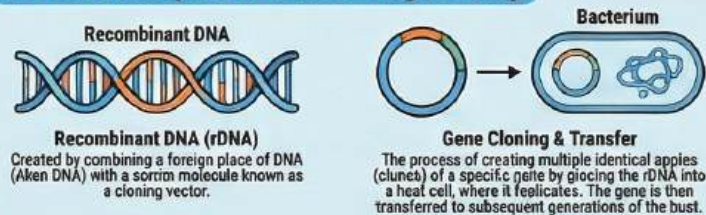
Biofertilizers: Enriching the Soil Naturally

Microorganisms that enrich the nutrient quality of the soil, increasing crop productivity without chemical fertilizers.

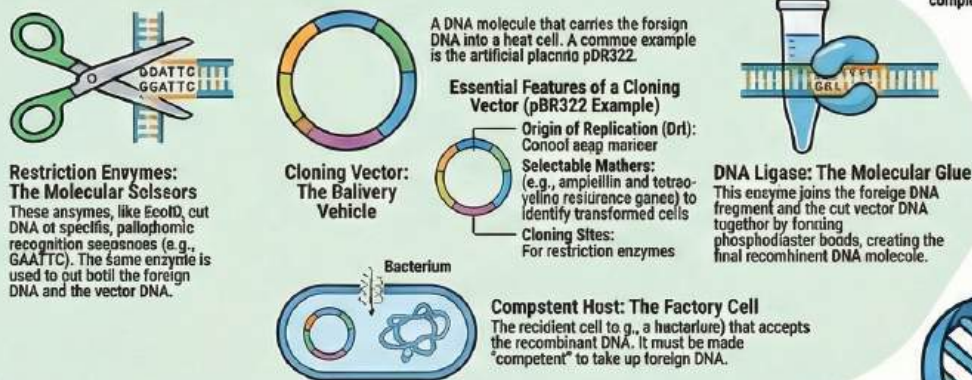
Bacteria (Symbiotic)	Nitrogen Fixation in Legumes	<i>Bradyrhizobium</i>
Bacteria (Free-Living)	Nitrogen Fixation in Soil	<i>Azospirillum</i> , <i>Azotobacter</i>
Fungi	Phosphorus Absorption for Plants (e.g., Glomus)	Mycorrhizas
Cyanobacteria	Nitrogen Fixation	Anabaena, Nostoc, Oscillatoria

The Blueprint of Biotechnology: A Guide to Recombinant DNA Technology

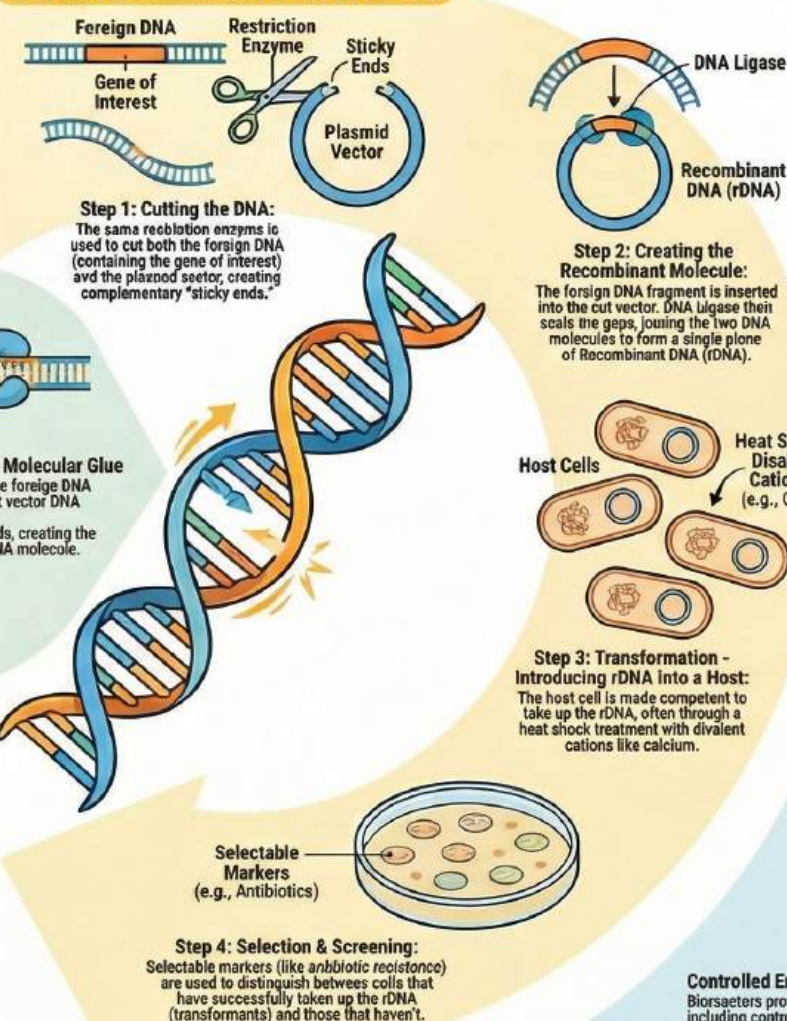
1. Core Principles of Genetic Engineering



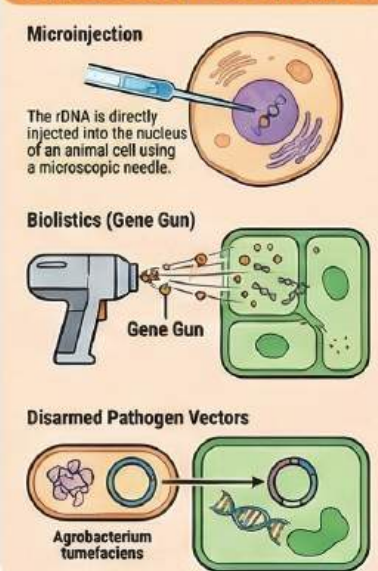
2. The Genetic Engineer's Toolkit



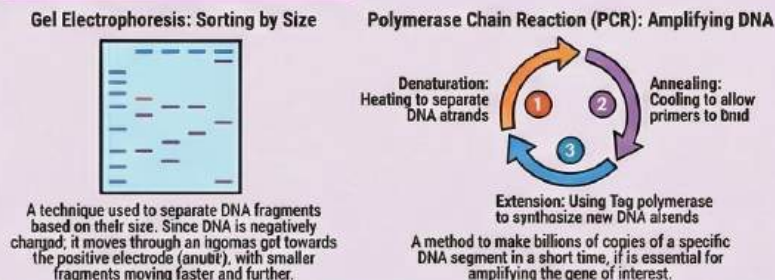
3. The Recombinant DNA Process



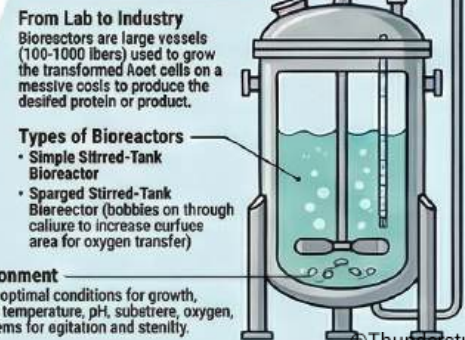
4. Methods of Gene Transfer



5. Amplification & Analysis



6. Large-Scale Production: The Bioreactor



A Visual Guide to Biotechnology and Its Applications

APPLICATIONS IN AGRICULTURE

GENETICALLY MODIFIED (GM) CROPS

Plants whose DNA has been altered using genetic engineering techniques to introduce new, desirable traits.

KEY ADVANTAGES

Increased Tolerance
(to abiotic stresses like drought, temperature)

Decreased Post-Harvest Losses

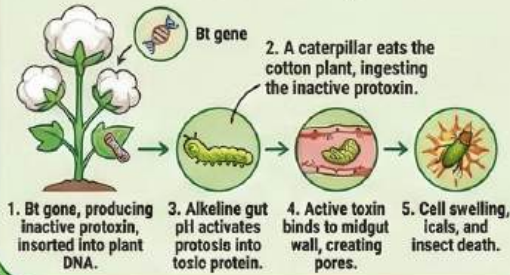
Reduced Reliance on Chemical Pesticides

Enhanced Nutritional Value

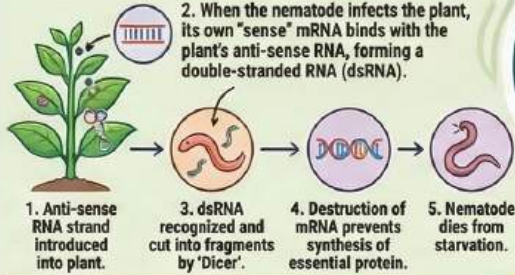
GOLDEN RICE

A GM crop enriched with β -carotenes, a precursor to Vitamin A, designed to combat vitamin deficiency.

HOW Bt COTTON WORKS (Pest-Resistant Cotton)



HOW RNA INTERFERENCE WORKS (Pest-Resistant Tobacco)



APPLICATIONS IN MEDICINE

GENETICALLY ENGINEERED INSULIN (HUMULIN)

OLD METHOD



Extracted from slaughtered animals (allergic reactions in humans)

NEW METHOD



Eli Lilly separately produced A and B chains and joined them (safe, non-allergenic human insulin)

GENE THERAPY

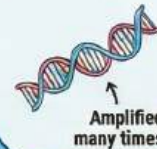
A technique used to correct a defective gene by replacing or inactivating it with a functional gene through genetic manipulation.



1990: FIRST CLINICAL USE
First-ever gene therapy on a 4-year-old for ADA deficiency.

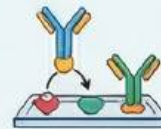
MOLECULAR DIAGNOSTICS FOR EARLY DISEASE DETECTION

Techniques that allow for identification of diseases before severe symptoms.



PCR (Polymerase Chain Reaction)

Amplifies minute DNA quantities, detects viruses/bacteria at low levels (e.g., RT-PCR for COVID-19).



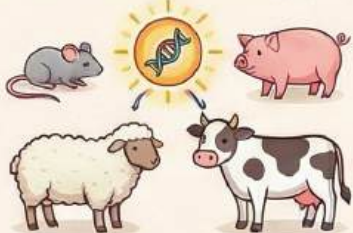
ELISA (Enzyme-Linked Immunosorbent Assay)

Detects presence of specific antigens or antibodies (e.g., for AIDS diagnosis).

TRANSGENIC ANIMALS

WHAT ARE TRANSGENIC ANIMALS?

Animals whose DNA has been manipulated to possess and express an extra, foreign gene.



MAJOR USES AND ADVANTAGES

Studying Disease Development

Producing Biological Products

Testing Safety (vaccines, chemicals)

ROSIE THE COW

A transgenic cow engineered to produce milk containing human α -lactalbumin (more nutritionally balanced for human infants).

ETHICAL & REGULATORY FRAMEWORK

GEAC (GENETIC ENGINEERING APPROVAL COMMITTEE)

Indian governmental body that makes decisions regarding the validity of GM research and safety of introducing GM organisms.

BIOPIRACY

Use of bio-resources by multinational companies without proper authorization and compensation.



BIOPATENT

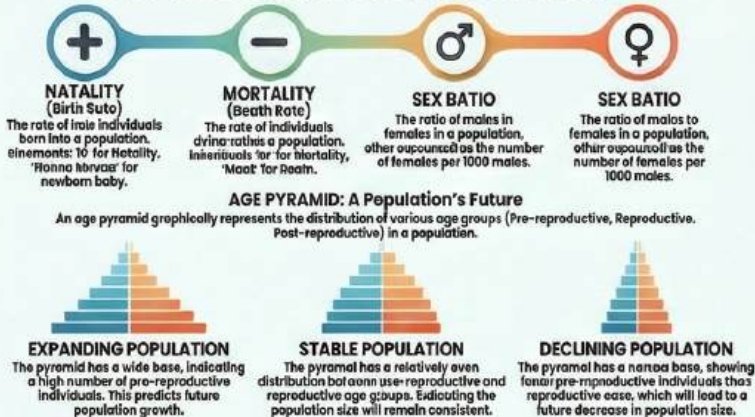
A legal patent granted for biological entities and products, giving the inventor exclusive rights (e.g., Basmati rice).



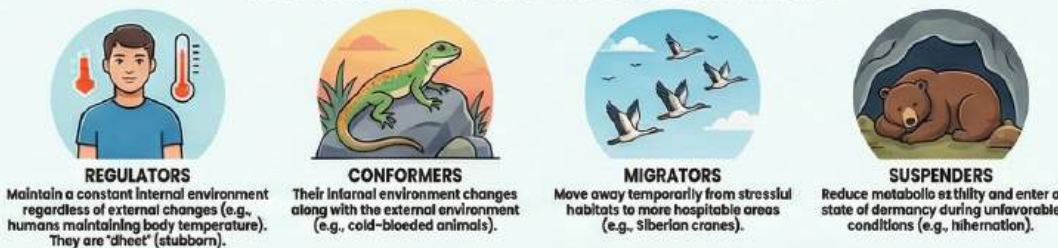
Organisms & Populations: A Quick Revision Guide



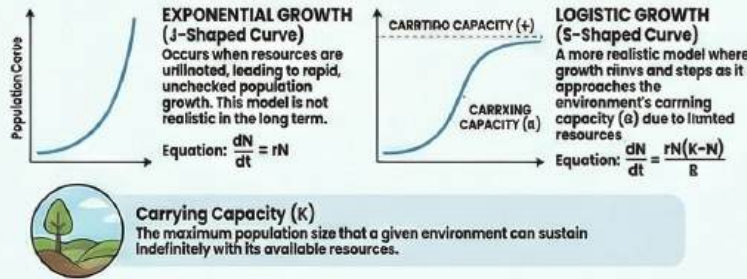
POPULATION ATTRIBUTES: THE VITAL SIGNS



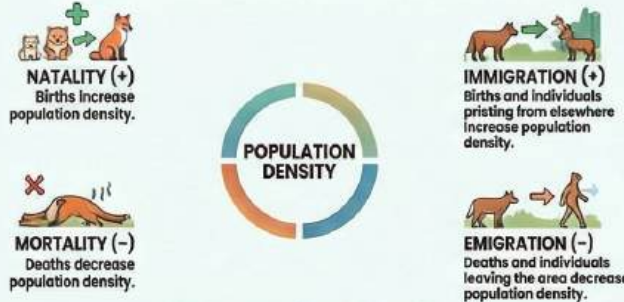
ORGANISM & ENVIRONMENT: SURVIVAL STRATEGIES



POPULATION GROWTH MODELS



FACTORS AFFECTING POPULATION DENSITY



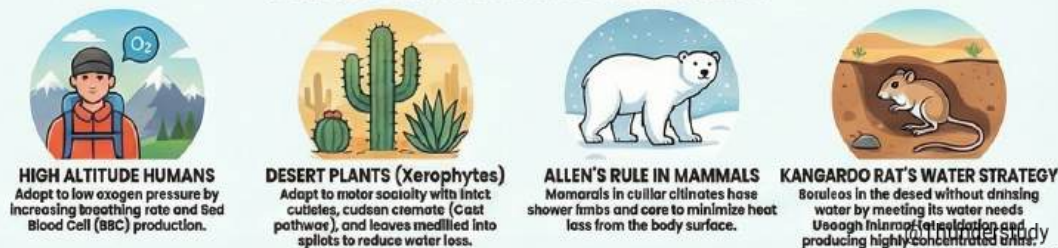
$$N(t+1) = N(t) + [(B + I) - (D + E)]$$

where B=Births, I=Immigration, D=Deaths, E=Emigration.

POPULATION INTERACTIONS: HOW SPECIES COEXIST

Interaction Type	Species A	Species B	Description
Mutualism	+	+	Both species benefit from the interaction.
Competition	-	-	Both species are harmed as they compete for the same resources.
Predation	+	-	One species (predator) benefits by killing and eating the other (prey).
Parasitism	+	-	One species (parasite) benefits at the expense of the other (host).
Commensalism	+	0	One species benefits, while the other is unaffected (neither harmed nor helped).
Amensalism	-	0	One species is harmed, while the other is unaffected.

ADAPTATIONS TO SPECIFIC ENVIRONMENTS



Ecosystem Essentials: A Visual Guide

Two Major Types of Ecosystems



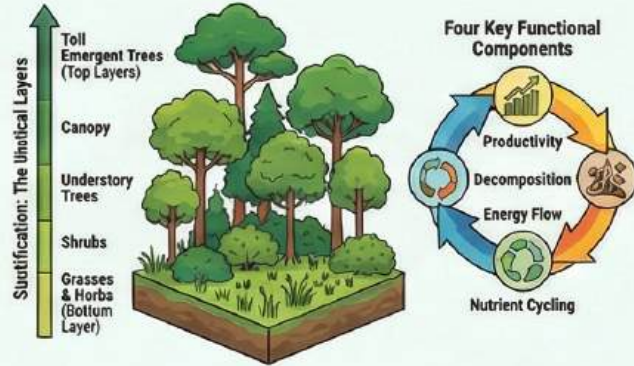
An ecosystem is a system formed by the interaction and integration of biotic (living) and abiotic (non-living) components.

Biotic Components

Abiotic Components

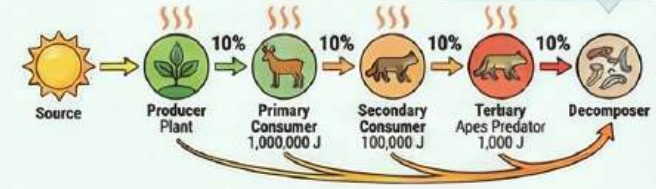


Ecosystem Structure & Function



Energy Flow & Food Chains

Energy Flow: The 10% Law

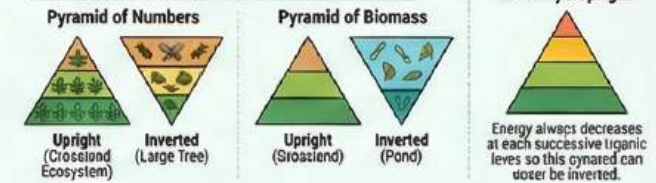


Two Types of Food Chains



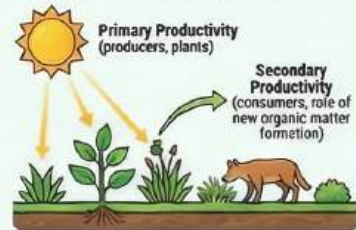
Ecological Pyramids: Visualizing Trophic Levels

Graphical representations of the number of individuals, biomass, or energy at successive trophic levels.

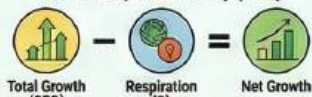


Productivity & Decomposition

Productivity: Creating Organic Matter



Net Primary Productivity (NPP)



Decomposition: The Ultimate Recycling

The process where detritivores break down dead organic matter (detritus) into simple inorganic substances.

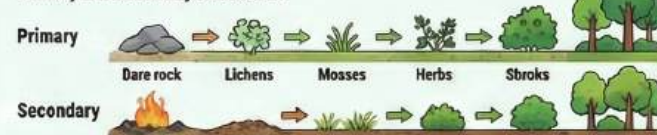


The 5 Steps of Decomposition

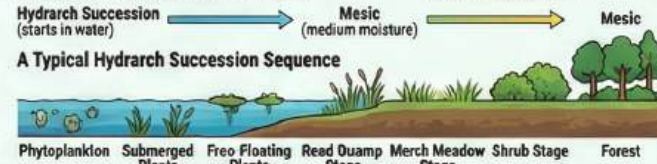


Ecological Succession

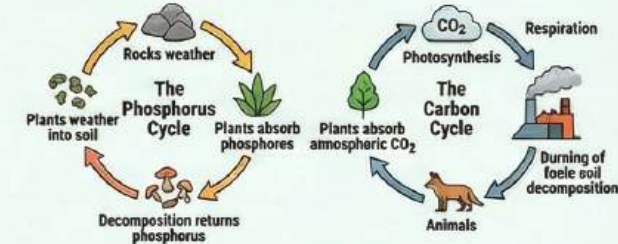
Primary vs. Secondary Succession



Succession in Different Environments



Nutrient Cycles & Ecosystem Services



Vital Services Provided by Ecosystems



A Visual Guide to Biodiversity & Conservation

THE THREE LEVELS OF BIODIVERSITY



Genetic Diversity: The variation of genes within a single species, which leads to different varieties.

Species Diversity: The variety of different species present within a specific geographical area.

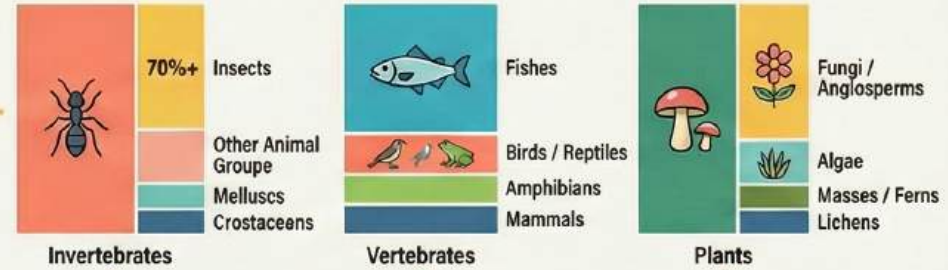
Ecological Diversity: The variety of different ecosystems, such as deserts, rainforests, and wetlands, in a region.

PATTERNS OF BIODIVERSITY

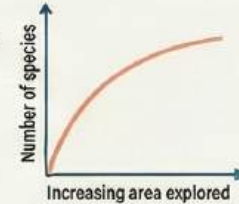


The Latitudinal Gradient: Species diversity decreases as you move from the equator (low latitude) towards the poles (high latitude). The tropics harbor more species than temperate or polar regions.

A SNAPSHOT OF GLOBAL SPECIES DIVERSITY



The Species-Area Relationship: As the explored area increases, the number of species found also increases, but only up to a certain limit.



$$S = CA^Z$$

S = Species richness
 A = Area
 Z = regression coefficient
 C = Y-intercept

THE "EVIL QUARTET": MAJOR CAUSES OF BIODIVERSITY LOSS



1. Habitat Loss and Fragmentation: This is the primary driver of extinction, caused by deforestation, urbanization, and creating barriers like roads that break up natural habitats.

2. Over-exploitation: Harvesting natural resources (like fish or trees) at a rate faster than they can recover, driven by human need and greed.

3. Alien Species Invasions: When a non-native species is introduced, it can lack natural predators and outcompete native species for resources, leading to their decline.

4. Co-extinctions: The extinction of one species leads to the extinction of another species that was dependent on it, like a specific plant and its pollinator.

WHY AND HOW WE MUST CONSERVE BIODIVERSITY

Why? The Utilitarian & Ethical Arguments



We conserve for direct benefits like drugs (28% are plant-derived), ecosystem services (oxygen, pollination), aesthetic pleasure, and because it is our moral duty.

How? Conservation Strategies



In-situ (On-site) Conservation
Protecting endangered species in their natural habitat. Examples include National Parks, Sanctuaries, Biosphere Reserves, and Sacred Groves.



Ex-situ (Off-site) Conservation
Protecting endangered species outside their natural habitat in controlled environments. Examples include Zoological Parks, Botanical Gardens, and Gene Banks.



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